A small thermal utility.
A measurable public return.

111 8th Avenue, New York City • A conditional investment case, an operating design and a working spatial decision tool.

12.50 GWh
Useful heat delivered / year

2,127 t
Net operational CO_2e avoided / year

$9.07m
Installed project capital

Choose urban density. Prove the paying demand first.

Select **111 8th Avenue** for the first project. A proposed 800 m shared network links compatible commercial hot-water and hydronic services with housing domestic hot water and a community facility. The model captures a 2.5 MW thermal tranche; it does not infer the whole facility's heat from its IT capacity. [S02; A: U_source_mw, U_route_m]

CAPTURE
30 → 22 °C
Isolated source loop

→

UPGRADE
3.0 MW
Modular heat pumps

→

DISTRIBUTE
65/40 °C
Insulated water pair

→

USE
≥63 °C
Metered building HX

The base case produces $874k annual operating cash, 10.4-year simple payback and $0.90m NPV over 25 years at 7%. Lake Hawkeye's modeled residential network has -$12.52m NPV. Dense, high-value demand outweighs the suburban case's warmer assumed source. These are scenario outputs, not observed project returns. [M: results.sites.U/R]

The decision today is diligence, not construction.
Commit $250k over 90 days to meters, routes and contracts. The urban case fails if an affordable electricity agreement, high-value anchor heat, sufficient contract duration or safe capture rights cannot be secured.

Evidence key throughout: S = source in the supplied folder; A = explicit registered assumption with range in the appendix/model; M = calculated model result. All prices are real 2026 USD. No site telemetry, customer commitments, vendor quotes or permits have been supplied. The brief governs scope; the Hacker Toolkit structures the business and pitch. [S02; S28 pp7,9,11]

Warmer heat is not enough.

Common physical and financial boundaries; different declared local-demand scenarios. Neither is a surveyed network.

Measure	111 8th Avenue — selected	Lake Hawkeye — defer
Source context	Commissioned, multi-tenant carrier hotel	Proposed former power-plant development
Capture water / tranche	30 °C / 2.5 MW	45 °C / 2.5 MW
Heat pumps / network	3.0 MW / 800 m	1.5 MW / 3,500 m
Demand / delivered heat	12.75 / 12.50 GWh/y	5.35 / 5.24 GWh/y
HP COP / system COP	3.68 / 3.47	5.18 / 3.95
Pipe heat loss	1.75%	16.99%
CAPEX / annual OPEX	$9.07m / $1.08m	$11.71m / $0.51m
LCOH / realized tariff	$149.8 / $156.0	$290.1 / $85.0
Annual operating cash	$874k	-$60k
NPV / simple payback	$0.90m / 10.4 y	-$12.52m / none
Net carbon reduction	2,127 t/y	841 t/y

Lake Hawkeye: test 300 compatible homes with space heat and DHW, plus 400 MWh/y community demand. Proposed liquid cooling can supply warmer water, but the 3.5 km trench, dispersed interfaces and winter-heavy loads overwhelm the $85/MWh affordable tariff. The DC must exist before customers can depend on it. [S02, S07; A: R_*]

Retain a suburban option for a co-located anchor or a future ambient loop with building heat pumps. Neither is included in this hot-network comparison. Any revised concept needs new costs and demand evidence. Residents must opt in, receive a verified bill comparison and retain full backup; anticipated acceptance is not evidence of consent. [S05 pp8–10; S19 pp7,24–26]

LCOH and tariff are $/MWh useful customer heat. Seasonal COP is energy-weighted. Source capacity, customer counts, routes and performance are assumptions. Urban and rural context is sourced; neither site's capture temperature is measured. Network loss denominator: heat entering the pipes. [M: results; A: inputs.json]

Contract the summer load.

Customer desirability means a compatible service, a credible avoided bill and a reason to stay connected.

Prospective urban cohort	Annual requirement	Why it belongs
Commercial anchor portfolio Hotels, kitchens/laundry preheat and compatible hydronic services	6,500 MWh DHW/process + 3,500 MWh space heat	Year-round hot-water demand supports utilization; higher avoided steam/electric variable cost can fund the network.
Housing DHW 900 target homes	2.5 MWh/home-year = 2,250 MWh	Steady hot water; resident savings covenant. Existing space heat remains outside the urban scope.
Community facility	200 MWh DHW + 300 MWh space heat	A metered public-serving connection with discounted heat and retained independent backup.

The search area is the immediate blocks around 111 8th Avenue. Candidate types and the 450 m farthest-branch assumption define a prospecting envelope, not a claim that named buildings are willing buyers. Verify plant rooms, water-side temperatures, trench rights and hourly meters before design. Housing unit targets are not demographic counts. [S02, S08; A: U_farthest_route_m]

The anchor is the hard part.
Commercial DHW/process demand alone implies roughly 340 thousand litres/day heated from 15 to 60 °C. This requires multiple real anchors, not a casual claim about one market or hotel. Remove that cohort and NPV falls to -$11.57m. [M: results; sensitivity]

Choose loads that the water can actually serve.

A 65 °C network supports 60 °C DHW production through a dedicated exchanger and compatible lower-temperature emitters. It cannot supply steam radiators or directly replace a steam main. The interface allowance covers suitable water-side conversions only. Local sanitary controls and booster capacity remain building responsibilities. [A: P_dhw, P_hx_approach, U_retrofit_cost]

Pools are low-temperature candidates but require humidity/ventilation work beyond pool water heat. Greenhouses have land and crop constraints and can need little summer heat. A 90 °C industrial process fails this network's temperature gate. All three are visible alternatives in the software and excluded from base revenue. [S05 pp8–9; S19 pp36–41; A: experience.json]

Profile assumptions are synthetic and normalized, not measured schedules. Commercial demand range and all cohort ranges appear in assumptions.csv. Early adoption targets building owners with central water systems, observable high variable heat costs and control over service contracts. [S28 pp7,9]

Three water boundaries. One protected cooling service.

Specify the capture node before discussing recoverable heat.

CAPTURE	UPGRADE	DISTRIBUTE	USE
30 → 22 °C	**3.0 MW**	**65/40 °C**	**≥63 °C**
Isolated source loop	Modular heat pumps	Insulated water pair	Metered building HX

Identify a specific existing condenser-water circuit upstream of ordinary heat rejection, with an authorized owner/operator. The urban assumption is 30 °C entering the capture exchanger and 22 °C leaving at nominal duty; inlet varies 25–35 °C in the annual model. Only a 2.5 MW tranche is assumed for screening. Winter free cooling and tenant circuit ownership can invalidate this assumption. [S15 p42; A: U_source_*]

Boundary / asset	Design duty and operating rule
DC → recovery interface	A plate exchanger separates water quality and pressure. Recovery sidestream closes on fault; the original full-duty cooling path remains operable. No IT temperature setpoint is changed.
Recovery → district water	3 modular heat pumps total 3.0 MW heating. COP derives from source/sink temperatures and a bounded Carnot-efficiency assumption. It is not a vendor map.
District → building water	65/40 °C design; at least 63 °C network water at the remote interface. Each ETS meters useful heat. Low-load flow increases to retain temperature; actual return can rise.
Distribution and buffer	800 m total twin-pipe trench, 450 m farthest route, about 28.7 kg/s design flow and 200 mm screening bore. 150 m^3 stratified storage.

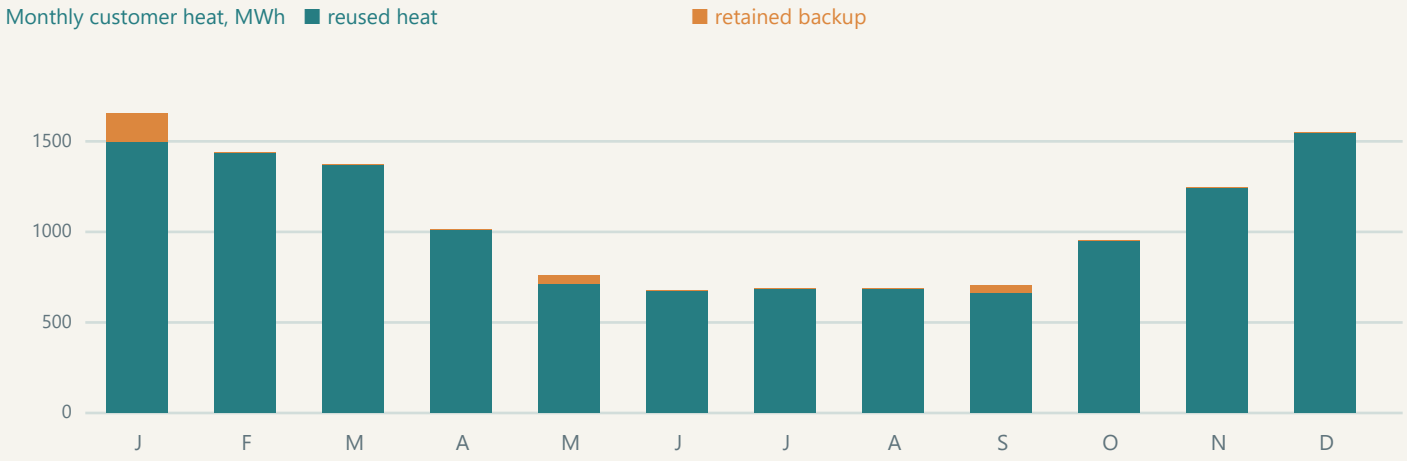
Keep the energy accounting honest.

Recovered source heat 9.293 GWh + compressor input 3.462 GWh = heat-pump output 12.755 GWh. Pipe loss 222 MWh and storage loss 32 MWh leave 12.501 GWh metered useful heat. Pumping, controls and sanitary boost bring new electricity to 3.598 GWh. [M: results.U]

Captured heat is not all heat-pump output: compressor work contributes useful heat. COPheating = QHP/Wcompressor; COPsystem = Qcustomer/(compressor + pumps + auxiliaries + sanitary boost). Seasonal values: 3.68 and 3.47. Design pipe bore is only a hydraulic screen; survey, velocities, pressure, freeze/corrosion protection and certified performance remain engineering gates. [S06 pp13–15; S17 s26,40; A: P_*]

Match every hour. Reject the heat nobody needs.

The reproducible model balances two synthetic 8,760-hour calendars and closes storage cyclically.



98.0%
Annual useful demand met by reuse

3.08 MW
Modeled peak customer demand

249 MWh
Annual heat from retained backup

Declared daily hot-water peaks and a mild DHW seasonal pattern combine with degree-hour space heating. Profiles normalize to 12.75 GWh urban demand. Source temperature also changes seasonally. The baseline includes 168 hours of export unavailability; source uptime is not inferred from IT uptime. [A: weather/profile parameters; model.py]

Operating state	Dispatch and continuity
Normal / summer	Recover only what customers can use and the permitted source can supply. Pro-rata hourly allocation across cohorts. Reject unused DC heat through ordinary cooling; never create demand to inflate reuse.
Winter peak	Available storage can discharge; building backup fills unmet heat. 3.71 MWh usable storage covers at most 1.21 hours at peak when full. No minimum emergency reserve is modeled.
Export fault / maintenance	Recovery stops. Full local backup must cover each building's design load. 6 MW aggregate assumed readiness exceeds the 3.07 MW modeled peak backup duty, but does not prove any individual building is ready.

Only 45.1% of the assumed tranche's available heat is captured; whole-site recovery is unknown. Total trench drives losses. Marginal-cost curtailment is implemented (base inactive); tariff dispatch is not optimized. [M: results; S18 pp12–14]

The cooling operator always has the final say.

Independent local protection; no web application or optimization model in the critical cooling control path.

Non-negotiable architecture
Retain the complete original cooling duty and redundancy. DC pumps, chillers/towers and emergency controls remain DC-owned. Recovery uses approved hydraulic separation, isolation and bypass. Heat-pump electrical supply is noncritical, separately protected and shed before IT loads. No critical UPS or generator capacity is borrowed. [Design requirements; S05 pp6–10]

Failure or conflict	Automatic response / responsibility
District pump, power or demand loss	DC-owned permissive withdraws recovery; fail-position isolation preserves the original rejection path. HeatCo diagnoses its loop; customers use local backup.
High pressure, leak or bad sensor	Independent exchanger boundary limits propagation; local limits and leak detection isolate the export circuit. Redundant/validated sensor design is required before commissioning.
Communications or software failure	Hardwired/local controls continue. Website and future supervisory software have no authority to block cooling or defeat safety interlocks.
Source maintenance or tenant change	Operator may curtail extraction. Advance notice where practicable, metered settlement and funded termination rules protect HeatCo; buildings retain continuity equipment.
Cold spell / one HP module down	Remaining modular capacity and charged storage supply what they can; full customer backup covers shortfall. No claim that the heat-pump plant itself is N+1 at full peak.
Wider power/gas interruption	Not proven by a source-outage simulation. Each building must validate emergency controls/pumps, available backup energy and a site emergency plan.

Prove safe failure before connecting a customer.

The independent engineer and DC operator witness loss of export, pump trip, valve fail-position, sensor error, pressure/leak event, communications loss, grid loss and planned maintenance. Acceptance requires the existing cooling service to remain inside the owner-approved thermal and hydraulic envelope in every test. Retrofit work uses approved isolation windows and temporary protection where required. [Proposed commissioning scope]

Heat continuity depends on verified building-level backup and enforceable maintenance duties, not by calling the tank “resilience.” Existing sanitary-water equipment remains responsible for safe DHW production and distribution. Storage may be empty when a fault occurs. [A: backup capacity and storage model; S14 p6; S19 p38]

A narrow commercial pass.

HeatCo pays for the system; the case excludes grants, debt, tax credits, carbon-credit revenue and unmeasured cooling savings.

Installed capital	USD thousands	Annual operating cost	USD thousands
Heat pumps	1,800	Electricity	810
Capture / isolation	600	Fixed O&M, admin, insurance/access	222
Twin pipes and civils	2,000	Source royalty	19
Customer ETS + interface retrofits	1,200	Community fund	25
Storage + electrical + controls	820	Total OPEX	1,075
Retained backup readiness	150	Heat revenue	1,950
Engineering + contingency	2,497	Net operating cash	874
Total CAPEX	9,067	Year 15 refurbishment	630

Heat is sold at $175.5/MWh to commercial customers against an assumed $195/MWh avoidable steam/electric baseline; public-serving/housing heat costs $85/MWh against $100/MWh. Retained fixed/standby bills and backup costs stay with customers. No connection fee is assumed. The insurance/asset-access placeholder is 0.4% of CAPEX annually; actual insurance and rent need quotes and a re-run. [A: U_*baseline/discount]

8.09%
Unlevered project IRR

10.37 y
Simple payback

20.15 y
Discounted payback at 7%

Do not transfer the data-center tariff to a new heat-plant meter.
The modeled electricity load factor is 45.2%; the source's indicative tariff context assumes 65–90%. $225/MWh is therefore an unverified contract assumption. At $280/MWh, NPV is -$1.41m; at $300/MWh it is -$2.25m. Obtain an interval utility quote with demand charges. [S22 p146; A/M]

Discounted LCOH $149.82/MWh versus weighted tariff $155.97/MWh. 25-year analysis, 7% real discount rate, year-15 replacement; no terminal value. CAPEX ranges are explicit screening assumptions, not bids. Totals calculated before display rounding. [M: results; appendix A]

Allocate each risk to someone who can manage it.

Proposed commercial terms; no party has signed or endorsed this concept.

Party	Assets, money and operating responsibility
DC circuit owner / operator	Owns original cooling and controls the extraction envelope. Can trip export for safety; provides metering/access under agreement. Receives $18,586/y at the modeled $2/MWh-source royalty. No cooling saving is booked.
HeatCo / ESCO	Funds $9.07m, including interface work once; owns/maintains capture, heat pumps, tanks, pipes and ETS to the delivery meter. Pays electricity, royalty, O&M and fund. Bears construction, ordinary performance and demand risk within contracts.
Building owners / customers	Pay metered useful heat; verify compatible loads and avoidable bills. Own/maintain downstream retrofits, local distribution, sanitary controls and full backup. Grant access. Housing owners pass savings to residents under covenant.
Community representatives	Approve benefit-fund priorities, inspect annual accounts and participate in route design. Independent resident representation, accessible consultation, complaints and fair opt-out rights.
Utility / municipality / route owners	Approve actual service capacity, tariff, excavation, easements and reinstatement. No permit, spare electrical capacity or funding award is presumed.
Engineer / EPC / vendors	Deliver scoped bids, certified performance maps, safe design and witnessed acceptance tests. Warrant workmanship and measured performance; arrange repair/defect remedies.

Contract for an asset life, not a handshake.

The source agreement identifies the exact circuit, extraction envelope, curtailment rights, maintenance notices and closure/technology-change obligations. A 10-year source-only horizon has -$2.93m NPV. Require term and credit support that match the financed cash flow, a funded termination remedy, or a separately costed replacement source. Unpriced insurance is not a solution. [S22 p77; M: sensitivity]

Offtake contracts specify temperature/return limits, hourly subscribed load, customer meter boundary and a minimum paid commitment. A two-part capacity tariff may later secure revenue, but the base model includes only actual metered energy sales. Price indexation must preserve a customer discount to verified avoidable cost; if electricity and that cap cannot coexist, do not build. [Proposed terms]

HeatCo carries reinstatement/decommissioning duties and insurance obligations under final contracts; asset transfer, step-in rights and termination funding require negotiation. Rent beyond the combined insurance/access allowance and end-of-life cash costs remain unpriced; add them before FID. Avoided building emissions are assigned once to the heat service; the DC may claim only separately measured changes in its own energy. No double sale of the same environmental benefit.

Count the benefit. Name who receives it.

Whole-service operational carbon accounting includes the heat that retained backup still supplies.

3,802 t
Baseline heat-service CO_2e / year

1,675 t
Proposed heat-service CO_2e / year

2,127 t
Net reduction / year

Baseline commercial heat is an explicit 70% steam / 30% resistance-electric mix. Housing/community use gas heat at 85% efficiency. Added electricity contributes 1,590 t/y; retained backup 74.4 t/y and refrigerant allowance 10 t/y are included. NYC electricity uses the folder's indicative 0.442 t/MWh, not a newly verified grid factor. [A: P_gas_ef, P_steam_ef, U_grid_ef; S22 pp51–52,147]

A separate 600 t construction allowance implies 0.28 years of carbon payback, but is not a product-EPD lifecycle study. At the high-grid sensitivity, reduction remains 2,023 t/y; with all commercial baseline heat treated as steam it is 1,621 t/y. These vary the baseline consistently. [M: sensitivity; A: embodied/leakage]

Recipient / metric	Modeled annual value	Condition for realizing it
Commercial heat customers	$191,158	Verified variable baseline and compatible water-side demand; retained fixed bills are excluded.
Housing: 900 target homes	$33,123 total; $36.80/home	Binding 100% resident pass-through and annual audit. This is not a 15% cut in the whole household utility bill.
Community facility	$7,349	Discount applies to the recovered-heat portion only.
Community benefit fund	$25,002	$2/MWh useful heat; already deducted from HeatCo cash flow. Residents choose and audit spending.

Use paid, accessible resident workshops before route selection; publish noise, access and reinstatement plans; offer a named grievance route and independent mediation. Track households receiving actual credits, dollars passed through, complaints resolved and fund allocations. No forced connection or unsupported job, health, food-production or neighborhood-temperature claim. [Proposed governance; S08, S19 pp24–26, S23 pp3–7]

Resource optimization: variable-flow pumps, modular capacity, limited storage and reuse of retained backup avoid unnecessary new assets. Closed-loop initial fill/makeup, refrigerant risk, construction materials, land/plant space and local disruption still matter. Quantified water savings are zero in the model because no site water baseline was supplied. [S25 pp10–13]

A decision tool you can challenge.

Built for the thermal developer, DC operator and community partner—not a replacement for safety controls.

Explore → trace → stress → decide

The interactive district is a native 3D scene with orbit, pan, zoom, cutaway server rooms, cooling equipment, storage, selectable applications, supply/return pipes and moving heat. Selecting a building traces its route and reveals the required load and temperature. Seasonal controls change the illustrative application duty; export-outage mode stops reuse and shows local backup.

The distance lab recalculates pipe temperature loss, required heat-pump output, electricity, circulation, pipe-only investment and variable heat cost. Insulation and flow determine temperature; there is no universal “heat dies at one kilometre” rule. Radial rings are distance guides, not atmospheric heat plumes. Rendered route geometry is measured separately; neither that scene nor its independent branch examples is the annual shared-network layout. [A: experience.json; model distance engine]

Explainable decision gate	What the software exposes
Source availability	Export outage gives zero recovered service; backup is required.
Temperature suitability	Delivered water must exceed the application requirement plus exchanger approach. A hot industrial process fails even beside the source.
Source and HP capacity	Required duty is compared with both finite recovery and heat-pump output capacity. A flagged requirement is not a delivery commitment.
Commercial confidence	Annual comparison exposes downside cases; input provenance, ranges and evidence gaps remain visible. Thermal feasibility alone does not approve investment.

Confidence is evidence plus sensitivity, not a fabricated accuracy score.
Only 13 of 27 declared combinations of electricity price, demand and capital cost have positive NPV. The grid is not a sampled population or a calibrated probability. All five field-evidence domains remain unvalidated. With no labeled site telemetry, an explainable physics/decision-tree model is more defensible than an invented trained ML model. [M: robustness.json; S29]

The same input register feeds an hourly Python model, generated browser solver, proposal and slide deck. Conservation/capacity tests and Python–JavaScript parity checks validate arithmetic. They do not validate the assumptions. Download the full inputs, hourly data and sensitivity results; replace assumptions with meters, bids and contracts to rerun the decision.

Browser branch results are steady-state required duties; variable cost excludes fixed overhead, CAPEX and local sanitary boost. Annual dispatch includes the boost allowance and backup. The software is a prototype without live SCADA, verified GIS or real customer data. [Model README; S10 pp149–177]

Desirable. Feasible. Conditionally viable.

Hacker's Canvas and Customer–Position–Revenue structure, adapted from the supplied toolkit. [S28 pp7,9,11]

Canvas field	Heat Commons
Problem / purpose	A data center rejects low-grade heat while nearby compatible services buy heat. Owners need a reliable, affordable contract—not a claim of free energy.
Customer / early adopter	Commercial building energy managers with central water circuits and high avoidable bills; housing and community owners with enforceable resident benefits. Location: compact urban catchment around 111 8th Avenue.
Existing alternatives	Retain steam/gas/electric heat; install individual heat pumps; use an ambient loop or co-locate a new anchor. Keep incumbent backup. These alternatives are the customer's negotiating benchmark.
Unique proposition / concept	An evidence-gated thermal utility: spatial heat matching plus transparent downside tests, protected cooling and an audited community dividend. The innovation is decision quality and accountable delivery; heat pumps are established technology.
Key benefits / metrics	Metered heat, carbon boundary, customer savings, resident pass-through, source royalty and fund. Track subscribed MWh, realized COP, tariff, operating cash, backup hours and safety-test acceptance.
Team talents / strengths	Six proposed functions: systems lead, thermal engineer, DC operations lead, model/software engineer, commercial lead and community lead. No fabricated hires, credentials or partner commitments.
Channels / education / network	Direct owner-to-owner source and anchor workshops; utility and route-owner design sessions; resident representatives. Use the 3D tool to explain temperature, distance, backup and bills before any subscription.
Cost to develop / revenue roadmap	$250k diligence; conditional $9.07m installation. Revenue is metered heat, not software licensing or carbon credits. Expand only when added routes clear the same evidence and financial gates.

CPR: the buying decision

Customer: the person accountable for heat cost, uptime and resident service. **Position:** a heat-as-a-service offer supported by measured inputs and a visible exit/no-build decision, with counterparties who control each asset. **Revenue:** measured useful heat at contracted tariffs; test affordability against actual avoidable bills, not total bills. [S28 p9]

Pitch follows desirability, feasibility and viability. When evidence fails, resize, change the portfolio or stop; model each alternative before booking its revenue. [S28 p11]

Spend on evidence before spending on pipe.

A 90-day, $250k proposed validation programme.

Work package / proposed lead	Budget	Required output
Source survey / DC + thermal leads	$60,000	Circuit rights, historical interval temperatures/flow and approved capture envelope; meter missing periods.
Customer audit / commercial lead	$50,000	Interval loads, plant surveys, avoidable bills and signed conditional subscriptions.
Route and design / systems lead	$70,000	Trench/utilities/easements, flood and access screen, structural plant-space approval, electrical tariff and scoped bids.
Contracts and equity / community + commercial leads	$40,000	Source/offtake terms, tenure/termination support, resident covenant and community governance.
Independent review / external engineer	$30,000	Cooling-protection review, model re-run, contingency and documented go/no-go recommendation.

Proceed only when these conditions hold together.

Source and safety: the measured permitted extraction envelope supports the hourly case; vendor maps and hydraulic design pass owner review. **Demand:** the base mix requires at least 11.87 GWh/y subscribed demand at unchanged tariffs, shapes and capital cost. That is a one-variable break-even threshold, not a substitute for a re-run.

Prices and cost: with all other base inputs fixed, electricity must remain below $246.35/MWh, and the commercial avoidable baseline must exceed $186.29/MWh at the proposed discount. Capital cost +20% yields -$1.32m NPV. Final jointly verified inputs must produce a financeable result; several individually passing thresholds can still fail together.

Contracts and consent: secure funded source/offtake continuity, affordable tariffs, resident pass-through, route rights, building backup and safety acceptance. If a critical gate fails, stop or model a smaller alternative. A short-source deal is not rescued by a long-life spreadsheet.

A practical sequence
Establish source rights and anchor bills; survey routes and obtain offers; consult residents, negotiate and rerun. Historical seasonal data are essential: this programme cannot observe a full year. Construction follows permits and owner-approved outages.

The $250k is pre-investment risk capital. If built it is credited inside the $985,500 engineering allowance; it is not added twice. If stopped, it is sunk diligence expenditure. Sources and assumption ranges are in the accompanying appendix; code and data permit reproduction.